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14. CONCLUSION

14.1 Conclusion of EIA Study

14.1.1 This EIA Report has provided an assessment of the potential environmental impacts associated with the construction and operation of the Project, based on the engineering design information available during the course of EIA study.

14.1.2 This EIA Report has been conducted, in accordance with EIAO-TM, covering the following environmental issues:-

- Air Quality Impact
- Noise Impact
- Water Quality Impact
- Waste Management Implications
- Land Contamination
- Ecological
- Fisheries
- Impact of Cultural Heritage
- Landscape and Visual Impact

14.1.3 The findings of this EIA study have determined the likely nature and extent of environmental impacts predicted to arise from the construction and operation of the Project. During the EIA process, environmental control measures have been identified and incorporated into the planning and design of the Project, to achieve compliance with environmental legislation and standards during both the construction and operation phases. The Implementation Schedule listing the recommended mitigation measures is presented in **Appendix 12.1**.

14.1.4 The various sections of this EIA Report have presented the measures to minimise the potential environmental impacts associated with the Project throughout the construction and operation phases. The key measures to minimise the environmental impacts are summarised in **Section 13**.

14.1.5 With the implementation of the mitigation measures, the Project would be environmentally acceptable with no unacceptable residual impacts on the environmentally sensitive receivers and resources. **Table 14.1** below summarises the key environmental outcomes and benefits that have accrued from the environmental considerations and analysis during the EIA process and the implementation of environmental control measures of the Project. **Appendix 14.1** presents the summary of environmental impacts and **Appendix 14.2** presents the summary of key assessment assumptions, limitation of assessment methodologies and related prior agreements with the Director.

14.1.6 An Environmental Monitoring and Audit (EM&A) programme has been developed to ascertain and verify the assumptions implicit to, and accuracy of, EIA study predictions.

EM&A requirements have been recommended, where necessary, to check on project compliance of environmental legislation and standards. These are presented in a separate stand-alone EM&A Manual.

Table 14.1 Summary of Key Environmental Outcomes and Benefits

Environmental Issue	Description
Environmental / Social Benefits of the Project	• **Northern Metropolis (NM) Synergy** – According to Northern Metropolis Action Agenda, the Hong Kong-Shenzhen Innovation and Technology Park (the Park) in the Loop is identified as a key area in innovation and technology zone, and KTN NDA is identified as a key area for providing housing supply with more than 80,000 housing units in NM. This Project will act as a key linkage linking up these two major development nodes. • **Enhance Traffic Connectivity** – Currently, the Western Connection Road is the only external transport link connecting the Loop and the existing road networks. The proposed ECR can share the traffic demand for the Western Connection Road, and can strengthen the connectivity of the Loop and the road networks within KTN NDA via Ma Tso Lung area.
Environmentally Friendly Design and Options	• By adopting the preferred alignment of the proposed ECR which incorporates depressed roads and underpasses, the project minimises disruption to critical habitats including wetlands, fishponds, and flight corridors for protected species like the Eurasian Otter and migratory birds. The use of natural ventilation for underpasses eliminates the need for energy-intensive mechanical systems, reducing long-term carbon emissions and visual impact. Additionally, the viaduct sections avoided direct impact to Shun Yee San Tsuen. During the operation phase, the ECR alleviates congestion on the WCR, lowering vehicular emissions from idling traffic and shortening travel times and results in further reducing fuel consumption and air quality impact.
Air Quality Impact	• With the implementation of mitigation measures specified in the Air Pollution Control (Construction Dust) Regulation together with the recommended air quality mitigation measures, good site practices, and EM&A programme, no adverse air quality impact would be anticipated during the construction phase of the Project. • The predicted NO_2, RSP and FSP concentrations at all existing and planned air sensitive receivers (ASRs) would comply with the prevailing Air Quality Objectives (AQOs). No adverse air quality impact would be anticipated during the operation phase of the project.
Noise Impact	• With the implementation of the proposed mitigation measures, construction noise impact to the NSRs could be effectively controlled and minimised. A construction noise management plan (CNMP) would be required to submit to the Director of the EPD. The CNMP will contain quantitative construction noise impact assessment, adopted quieter construction method and

Environmental Issue	Description
	equipment, noise mitigation measures, and construction noise monitoring and auditing programme, with reference to the updated and identified noise mitigation measures once available and in any case before the tender invitation if there is any change to the construction noise mitigation measures recommended in this EIA report and before the commencement of construction of this Project. Any technical constraint that would hinder the use of these quieter construction method and equipment shall be evaluated and clearly recorded in the assessment. • While for the road traffic noise, the assessment results indicate that the predicted road traffic noise levels at some of the representative NSRs would exceed the noise criteria under the unmitigated scenario. Direct noise mitigation measures have been considered to alleviate the potential traffic noise impact. With implementation of the recommended noise mitigation measures of LNRS and the provision of at-receiver mitigation measures, such as acoustic windows/balconies, the noise levels at all NSRs within the assessment area would comply with the traffic noise criteria stipulated in Clause 4.7.6 of EIAO GN No. 12/2023. As such, adverse residual road traffic noise impact due to the Project is not anticipated. • Since the design of pumping station is fully enclosed with the exhaust louver oriented to the north, facing Hoo Hok Wai. This configuration is intended to minimise the impact of potential fixed noise sources. No residual impact would be expected from fixed noise source with proper implementation of the recommended mitigation measures.
Water Quality Impact	• The construction of underpass road across the LMC Meander would require installation of cofferdam to enclose the excavation works. The construction of underpass road across the LMC Meander should be conducted by phases and only one cofferdam should be installed in each stage. Cofferdams should be designed by the Contractor which fulfil the requirements in DSD Technical Circular No. 14/2017 "*Temporary Flow Diversions and Temporary Works Affecting Capacity in Stormwater System*". Water pumps should be used to collect any construction site runoff and ingress/seepage water within the cofferdam to the on-site wastewater treatment facilities for treatment to satisfactory levels before discharge. Silt curtain should be deployed to completely enclose the cofferdam prior to conduct cut-and-cover construction works in the LMC Meander, and installation and removal of cofferdams. • Water quality impacts from land-based construction would be controlled by implementing the recommended mitigation measures and good site practices. Water quality monitoring and regular site inspections would be carried out routinely to minimise water quality impacts upon the nearby inland waters. An Emergency Response Plan is recommended to minimise the potential water quality impact from construction site

Environmental Issue	Description
	discharges under failure of treatment facilities during emergency situations or inclement weather. • Non-point source surface runoff generated from the additional paved areas was identified as potential water quality impact. The associated water quality impact was expected to be minimal and acceptable, provided that the recommended mitigation measures for the drainage system are properly implemented.
Waste Management Implications	• The main waste types generated during the construction of the Project would be C&D materials (from site clearance, excavation/foundation works, slope works, etc.), excavated sediment, chemical waste from plant operation and maintenance and general refuse from workforce. Provided that these wastes are handled, reused, transported and disposed of using approved methods and that the recommended good site practices are strictly followed, no adverse environmental impacts are anticipated. • During the construction of the Project, the quantity of C&D materials generated during the construction phase is estimated to be 313,100 m^3 with 5,900 m^3 of non-inert materials and 307,200 m^3 of inert materials. It is estimated that of the inert C&D materials generated, approximately 15,700 m^3 would be reused on site as backfill materials. Surplus inert C&D material will be exported off site or delivered to PFRF (Tuen Mun Area 38 Fill Bank). A minimal amount (in the order of a few hundred litres per month) of chemical waste and 65 kg of general refuse are estimated to be generated daily. Chemical waste would be recycled at an appropriate facility as far as possible, while chemical waste that cannot be recycled should be disposed of at CWTC, or another licensed facility. General refuse would be disposed of at WENT / NENT Landfill. • It is estimated that 100,700 m^3 of excavated sediment will be generated from the Project during underpass/depressed road excavation. Excavated sediment will be treated by cement S/S to improve its strength characteristics and following confirmatory UCS and TCLP testing, will be reused on site (e.g. as backfilling materials). With the implementation of the recommended mitigation measures detailed in Section 6.7, no adverse environmental impacts would be expected from excavation and handling of sediments. • No adverse environmental impact (including potential hazard, air and odour emissions, noise, wastewater discharge and public transport) is anticipated by the handling, collection, transportation and disposal of the different types of waste. • During operation of the Project, minimal amount of chemical waste and general refuse are estimated to be generated during each maintenance event.
Land Contamination	• The assessment assessed the potential for contaminative land use within the Project boundary. Based on the findings of desktop review (e.g. a review of historical aerial photographs

Environmental Issue	Description
	and relevant information from EPD and FSD) and a site walkover; no potential land contaminating land use was identified except for Site MTL-001. • It is recommended to re-appraise the whole area within the Project boundary once the Project Site is resumed and fully accessible prior to development. Further works, including site re-appraisal, SI works as well as submission of a CAP and CAR for EPD's endorsement are recommended to be carried out after the Project site is handed over to the PP for development. If contaminated soil and/or groundwater is identified; remediation should be carried out according to the EPD approved RAP and a RR should be submitted to EPD for agreement after completion of the remediation works. No construction or development works should be carried out on potentially contaminated site prior to EPD approval of the CAP, and CAR and/or RR, if necessary. • With reference to previous studies, natural occurrences of elevated level of arsenic exist over broad areas of the northern New Territories. A separate arsenic assessment and remediation plan should be prepared and submitted to EPD for endorsement. • With the implementation of the recommended further works, any soil/groundwater contamination would be identified and properly treated prior to the construction works. No insurmountable land contamination impacts to the Project are therefore anticipated.
Ecological Impact	• The potential ecological impacts arising from the construction works of the ECR would include habitat loss, disturbance to wetland, terrestrial habitats and species of conservation importance, potential obstruction of major flight paths, habitat fragmentation, mortality risks to species of conservation importance and other indirect impacts on sites of conservation importance, natural habitats and associated wildlife in the vicinity. With the implementation of avoidance, mitigation and compensation measures recommended in Ecological Impact Assessment, no unacceptable residual ecological impact is anticipated. These measures will ensure “no-net-loss” in wetland functions as a result of the development.
Fisheries	• The occupation or suspension of fisheries operation is of negligible significance to local aquaculture production. With the implementation of good site practices specified in the EIA, no unacceptable residual fisheries impacts is anticipated.
Impact of Cultural Heritage	• MacIntosh Fort (Ma Tso Lung) (HB303) (grade 2 historic building) is situated within 50m from the Site Boundary. Potential indirect impacts or ground borne vibration, tilting and settlement would be anticipated during the construction phase. Therefore, during construction phase, monitoring of ground-borne vibration, tilting and ground settlement is proposed to be employed incorporating with a set of 3As. Furthermore, an Authorized Person should be employed to carry out pre and

Environmental Issue	Description
	post condition survey and structural assessment during construction phase. • There is no site of archaeological interest within the Site Boundary and its 300m assessment area, and it is of no archaeological potential. No archaeological impact is anticipated.
Landscape and Visual	• Project Site Boundary, approx. 32% of existing trees (i.e. approx. 327 nos. of trees, including 4 nos. of TPIs (i.e. Aquilaria sinensis) and approx. 5 nos. of undesirable species (i.e. Leucaena leucocephala)) would be unavoidably affected by the proposed works. Transplantation of all 4 nos. of affected TPIs (Aquilaria sinensis (土沉香)) is recommended. Tree compensation within the site in a form of woodland mix tree planting on slopes or heavy standard tree planting in amenity area would be explored. Tree compensation requirement of 1:1 in terms of number should be carried out as far as practicable. Exact number of trees to be retained, transplanted and removed to be determined under the Tree Preservation and Removal Proposals (TPRPs) in accordance with DEVB TC(W) – No. 4/2020 - Tree Preservation. • Appropriate landscape and visual mitigation measures as below would be implemented: <u>Construction Phase:</u> • Trees without impact from the proposed works should be retained and any existing trees unavoidably affected by the works should be transplanted as far as possible in accordance with DEVB TC(W) No. 4/2020. • Erection of decorative screen hoarding or hoarding compatible with the surrounding setting. • Construction facilities and activities on work sites and areas should be carefully managed and controlled, including height and disposition / arrangement to minimise any potential adverse landscape impacts. • Reinstatement of the temporarily affected landscaped area within assessment area in like-for-like basis to restore the existing natural environment as far as possible. • Coordinated implementation programme with concurrent projects to minimise impacts and where possible reduce the period of disturbance. <u>Operational Phase:</u> • Aesthetically pleasing design as regard to the form, material and finishes should be incorporated into all aboveground structures, roads, viaducts, ancillary buildings and other associated engineering facilities so as to blend in the structures with the adjacent landscape and visual context. • Buffer screen planting, including shrubs, to provide screening. • As far as practicable, compensatory tree planting requirement of 1:1 in terms of number in accordance with DEVB TC(W) No. 4/2020 for compensation of felled trees subject to the

Environmental Issue	Description
	availability of space and agreement made with relevant departmental parties. • Temporary loss of fishponds along ECR by the road widening and improvement works will be largely reinstated. These ponds will be used for both functional or amenity purposes to enhance the existing landscape and visual context. • Landscape treatments on slope or retaining structure in accordance with GEO Publication No. 1/2011 – Technical Guidelines on Landscape Treatment for Slopes shall be adopted subject to the availability of space and agreement made with relevant departmental parties. <u>Conclusion:</u> • With the implementation of the proposed landscape and visual mitigation measures above, the overall landscape residual impacts would be from Negligible to Slight in operational phase, and the overall visual impacts would be from Negligible to Moderate in operational phase. With full implementation of the recommended mitigation measures, unacceptable adverse residual landscape impacts and visual impacts are not expected.